

Example 2.1

$$X_t = X_{t-1} + Z_t$$

(a) Show that this process is not (weakly) stationary.

We have $X_t - X_{t-1} = Z_t$

「 $\nabla X_t = Z_t$ 」

$$\Rightarrow (1-B)X_t = Z_t$$

$$\Rightarrow X_t = \frac{1}{1-B} Z_t$$

$$= \sum_{j=0}^{\infty} B^j Z_t$$

「Taylor series
exp. of
 $\frac{1}{1-B}$ 」

$$= \sum_{j=0}^{\infty} Z_{t-j}$$

$$\mathbb{E}(X_t) = \mathbb{E}\left[\sum_{j=0}^{\infty} Z_{t-j}\right]$$

$$= \sum_{j=0}^{\infty} \mathbb{E}(Z_{t-j})$$

$$= 0 \quad \because \mathbb{E}(Z_t) = 0 \text{ for all } t.$$

Consider

$$\text{Var}(X_t) = \text{Var}\left(\sum_{j=0}^{\infty} Z_{t-j}\right)$$

$$= \sum_{j=0}^{\infty} \text{Var}(Z_{t-j}) + \sum_{j \neq k} \text{Cov}(Z_{t-j}, Z_{t-k})$$

↗ 0

$$= \sum_{j=0}^{\infty} \text{Var}(Z_{t-j}) \quad \text{because } \{Z_t\} \text{ is a white noise process, } Z_{t-j}, Z_{t-k} \text{ are independent if } j \neq k.$$

$$= \sum_{j=0}^{\infty} \sigma_Z^2 \quad (\text{Var}(Z_t) = \sigma_Z^2 \text{ for all } t)$$

$$= \sigma_Z^2 \sum_{j=0}^{\infty} 1 = \infty$$

$\therefore$ Variance of the process $\{X_t\}$ is not finite

$\Rightarrow \{X_t\}$ is not a stationary process.

(b) We have $X_t = X_{t-1} + Z_t$

If we let $Y_t = X_t - X_{t-1} \quad (\forall X_t)$
 $= Z_t$

Then $\{Y_t\}$ is a white noise process and therefore stationary.